Volumes Of Revolution

Solutions

Chapter Overview

Find the volume of a solid when a curve is revolved around either coordinate axis.

1. Rotate around the x-axis.

Find the volume formed by rotating $y = x^3$ about the x-axis between $x = 1$ and $x = 2$.

2. Rotate around the y-axis.

Find the volume formed by rotating $y = \sqrt{x+1}$ about the y-axis between $y = 1$ and $y = 2$.

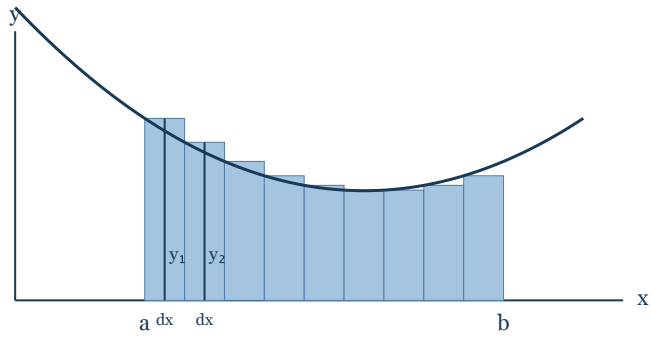
3. Find more complex volumes by adding and subtracting.

The area bounded by $y = \sqrt{x}$ and $y = x$ is rotated through $360°$ about the x-axis. Find the volume of the resulting solid.

Area under a Graph

Solution 3

For $f(x) \geq 0$, $\int_a^b y\,dx$ gives the area bounded by $y = f(x)$, $x = a$, $x = b$ and the x-axis. Why?



Split the area into thin rectangular strips. Each has width dx, height $y = f(x)$, and area $f(x)\,dx$.

For discrete strips, add their areas: $\sum f(x)\,\Delta x$.

As the widths tend to zero, use $\int$ instead of Σ:

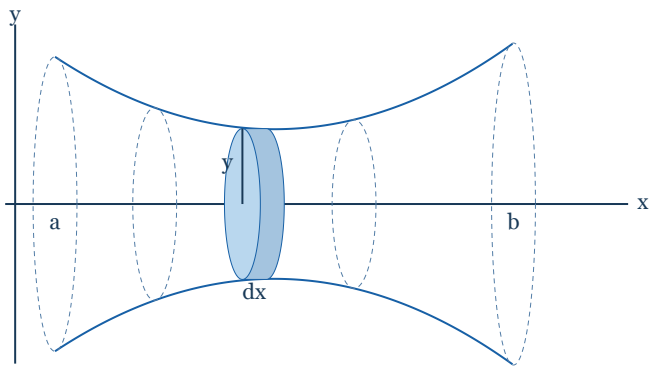
$$A = \int_a^b f(x)\,dx.$$

Integration is a continuous version of summation.

Volumes of Revolution

Solution 4

Spin $y = f(x)$ about the x-axis to form a solid of revolution. How can we use the same principle as before? What replaces strips?



Each thin cylinder has volume:

$$\pi y^2\,dx.$$

Add all the thin cylinders:

$$V = \pi \int_a^b y^2\,dx.$$

Square the function and put a factor of π in front.

Rotate the curve around the axis.

Use infinitely thin cylinders, each with width dx and radius y.

Example

Solution 5

[Textbook] The diagram shows the region R which is bounded by the x-axis, the y-axis and the curve with equation $y = 9 - x^2$. The region is rotated through 360° about the x-axis. Find the exact volume of the solid generated.

Working

Find the roots first:

$9 - x^2 = 0 \implies x = \pm 3.$

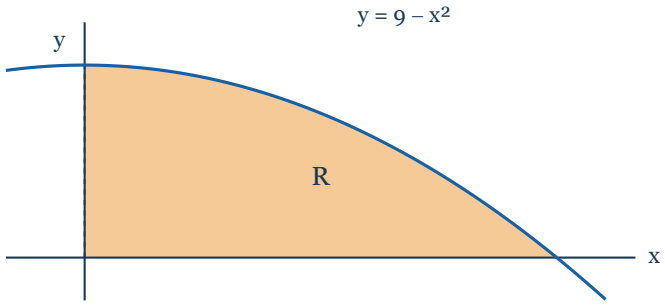
Use the positive region:

$$V = \pi \int_0^3 (9 - x^2)^2 \, dx$$

$$= \pi \int_0^3 (81 - 18x^2 + x^4) \, dx$$

$$= \pi \left[81x - 6x^3 + \tfrac{1}{5}x^5\right]_0^3$$

$$= \pi \left(\tfrac{648}{5} - 0\right) = \tfrac{648\pi}{5}.$$



Test Your Understanding

Solution 6

Edexcel C4 June 2013(R), Q7(c)

$C: \ y = \sqrt{x^{2/3} - 9}$. The finite region R is bounded by C, the x-axis and $x = 125$, as shown. Rotate R through 360° about the x-axis. Use calculus to find the exact volume. (5 marks)

Working

At the intercept, $x^{2/3} = 9$, so $x = 27$ on this branch.

$$V = \pi \int_{27}^{125} \left(\sqrt{x^{2/3} - 9}\right)^2 \, dx$$

$$= \pi \int_{27}^{125} (x^{2/3} - 9) \, dx$$

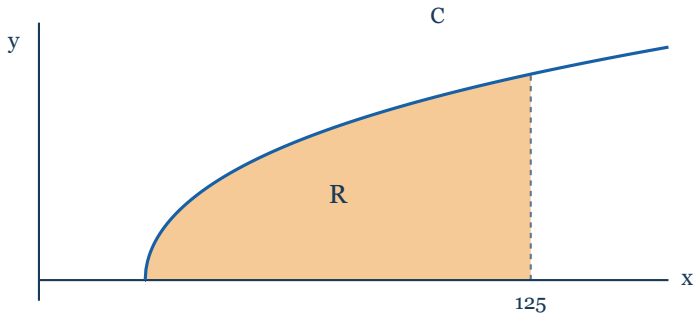
$$= \pi \left[\tfrac{3}{5}x^{5/3} - 9x\right]_{27}^{125}$$

$$= \pi\big((\tfrac{3}{5}(125)^{5/3} - 9(125))$$

$$- (\tfrac{3}{5}(27)^{5/3} - 9(27))\big)$$

$$= \pi\big((1875 - 1125) - (145.8 - 243)\big)$$

$$= \frac{4236\pi}{5} = 847.2\pi.$$



Exercise 5A — Core Pure Year 1 Pages 73–75 Solution 7

1. Find the exact volume when each curve is rotated through 360° about the x-axis between the given limits.

(a) $y = 10x^2,\ 0 \le x \le 2.$

a

$$V = \pi \int_0^2 (10x^2)^2\,dx$$
$$= \pi \int_0^2 100x^4\,dx$$
$$= \pi[20x^5]_0^2$$
$$= \pi(640 - 0) = 640\pi.$$

(b) $y = 5 - x,\ 3 \le x \le 5.$

b

$$V = \pi \int_3^5 (5-x)^2\,dx$$
$$= \pi \int_3^5 (25 - 10x + x^2)\,dx$$
$$= \pi[25x - 5x^2 + x^3/3]_3^5$$
$$= \pi((125 - 125 + 125/3)$$
$$-(75 - 45 + 9))$$
$$= \pi(125/3 - 39) = 8\pi/3.$$

(c) $y = \sqrt{x},\ 2 \le x \le 10.$

c

$$V = \pi \int_2^{10} (\sqrt{x})^2\,dx$$
$$= \pi \int_2^{10} x\,dx = \pi[x^2/2]_2^{10}$$
$$= \pi(100/2 - 4/2) = 48\pi.$$

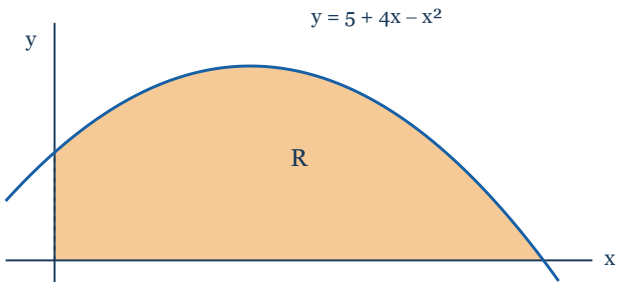
(d) $y = 1 + x^{-2},\ 1 \le x \le 2.$

d

$$V = \pi \int_1^2 (1 + x^{-2})^2\,dx$$
$$= \pi \int_1^2 (1 + 2x^{-2} + x^{-4})\,dx$$
$$= \pi[x - 2/x - 1/(3x^3)]_1^2$$
$$= \pi((2 - 1 - 1/24) - (1 - 2 - 1/3))$$
$$= \pi(23/24 + 4/3) = 55\pi/24.$$

Exercise 5A — Core Pure Year 1 Pages 73–75 Solution 8

2. The curve is $y = 5 + 4x - x^2$. The finite region R is bounded by the curve and both coordinate axes. Rotate R through 2π radians about the x-axis. Find the exact volume.



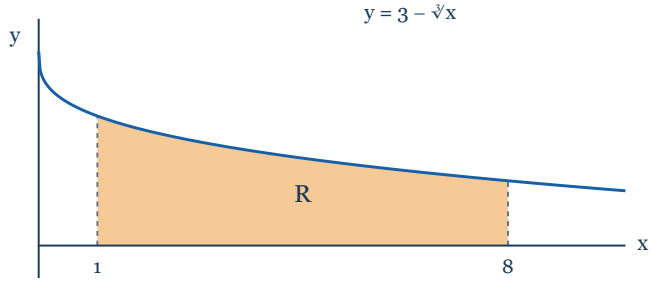
2

$5 + 4x - x^2 = (5 - x)(1 + x) = 0.$

Since $x > 0$, $x = 5$.

$$V = \pi \int_0^5 (5 + 4x - x^2)^2\,dx$$
$$= \pi \int_0^5 (25 + 40x + 6x^2 - 8x^3 + x^4)\,dx$$
$$= \pi[25x + 20x^2 + 2x^3 - 2x^4 + x^5/5]_0^5$$
$$= \pi(125 + 500 + 250 - 1250 + 625) = 250\pi.$$

3. R is bounded by the x-axis, $x = 1$, $x = 8$ and $y = 3 - \sqrt[3]{x}$. Rotate R through 2π radians about the x-axis. Find the exact volume.



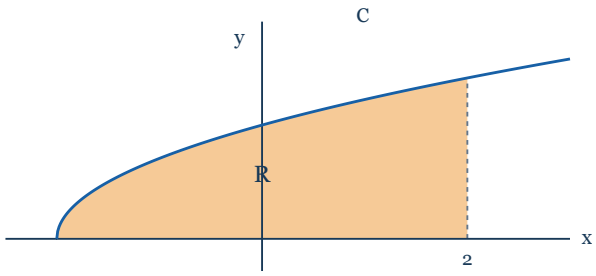
3

$$V = \pi \int_1^8 (3 - \sqrt[3]{x})^2\,dx$$
$$= \pi \int_1^8 (9 - 6x^{1/3} + x^{2/3})\,dx$$
$$= \pi[9x - \tfrac{9}{2}x^{4/3} + \tfrac{3}{5}x^{5/3}]_1^8$$
$$= \pi((72 - 72 + 96/5) - (9 - 9/2 + 3/5))$$
$$= \pi(96/5 - 51/10) = 141\pi/10.$$

Exercise 5A — Core Pure Year 1 Pages 73–75

Solution 9

4. Curve C has equation $y = \sqrt{x+2}$. Region R is bounded by C, the x-axis and $x = 2$. Rotate R through $360°$ about the x-axis. Find the exact volume.



4

$$\sqrt{x+2} = 0 \implies x = -2.$$

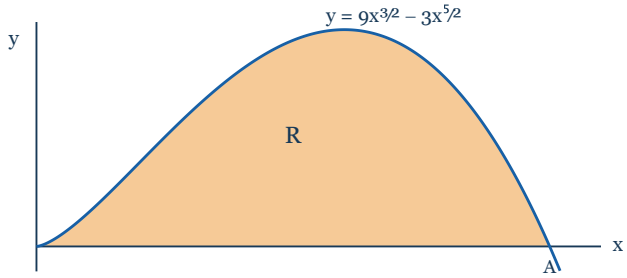
$$V = \pi \int_{-2}^{2} (\sqrt{x+2})^2 \, dx$$

$$= \pi \int_{-2}^{2} (x+2) \, dx$$

$$= \pi [x^2/2 + 2x]_{-2}^{2}$$

$$= \pi((2+4) - (2-4)) = 8\pi.$$

5. $y = 9x^{3/2} - 3x^{5/2}$. Region R is bounded by the curve and the x-axis. (a) Find the positive intercept A. (b) Rotate R through 2π radians about the x-axis; find the volume.



5a

$$3x^{3/2}(3-x) = 0 \implies x = 0 \text{ or } 3.$$

$$A = (3, 0).$$

5b

$$V = \pi \int_{0}^{3} (9x^{3/2} - 3x^{5/2})^2 \, dx$$

$$= \pi \int_{0}^{3} (81x^3 - 54x^4 + 9x^5) \, dx$$

$$= \pi [81x^4/4 - 54x^5/5 + 3x^6/2]_{0}^{3}$$

$$= \pi(6561/4 - 13122/5 + 2187/2)$$

$$= 2187\pi/20.$$

Exercise 5A — Core Pure Year 1 Pages 73–75

Solution 10

6. $C: \ y = \sqrt{3x^4 - 3}/x^3$. The positive region bounded by C, the x-axis and $x = 6$ is rotated through 2π radians about the x-axis. Find the volume to 3 significant figures.

Working

$$\frac{\sqrt{3x^4 - 3}}{x^3} = 0 \implies x = \pm 1.$$

The graph has $x > 0$, so use $x = 1$.

$$V = \pi \int_{1}^{6} \left(\frac{\sqrt{3x^4 - 3}}{x^3} \right)^2 dx$$

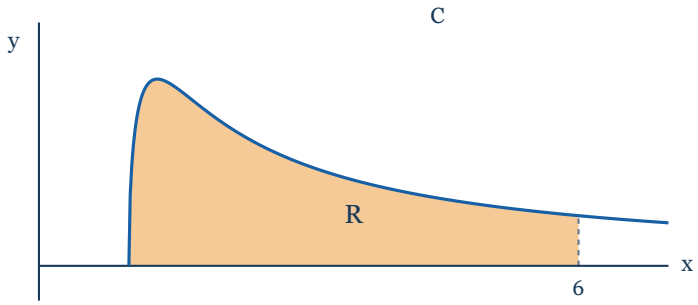
$$= \pi \int_{1}^{6} \frac{3x^4 - 3}{x^6} \, dx$$

$$= \pi \int_{1}^{6} (3x^{-2} - 3x^{-6}) \, dx$$

$$= \pi [-3/x + 3/(5x^5)]_{1}^{6}$$

$$= \pi((-3/6 + 3/38880) - (-3 + 3/5))$$

$$= 5.97 \quad (3 \text{ s.f.}).$$



Exercise 5A — Core Pure Year 1 Pages 73–75 Solution 11

7. The curve is $5y^2 - x^3 = 2x - 3$. The shaded region is bounded by the curve and $x = 4$. Rotate it about the x-axis. Find the volume.

Working

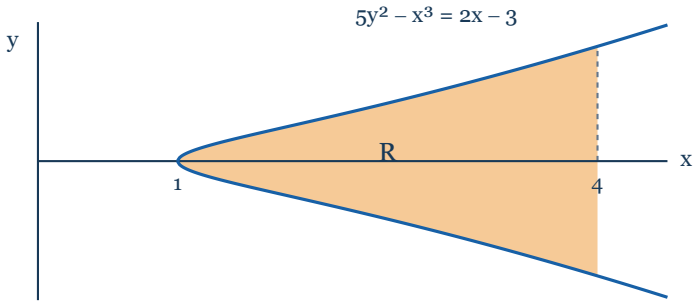
$$5y^2 - x^3 = 2x - 3 \implies y^2 = \tfrac{1}{5}(x^3 + 2x - 3).$$

$$V = \frac{\pi}{5}\int_1^4 (x^3 + 2x - 3)\,dx$$

$$= \frac{\pi}{5}[x^4/4 + x^2 - 3x]_1^4$$

$$= \frac{\pi}{5}((64 + 16 - 12) - (1/4 + 1 - 3))$$

$$= \frac{\pi}{5}(68 + 7/4) = 279\pi/20.$$



Exercise 5A — Core Pure Year 1 Pages 73–75 Solution 12

8. $y = x\sqrt{4 - x^2}$. Region R lies between the curve, the x-axis and $x = a$, where $0 < a < 2$, as shown. Rotation through 2π radians about the x-axis gives volume $657\pi/160$. Find a.

Working

$$V = \pi\int_a^2 (x\sqrt{4 - x^2})^2\,dx$$

$$= \pi\int_a^2 x^2(4 - x^2)\,dx$$

$$= \pi\int_a^2 (4x^2 - x^4)\,dx$$

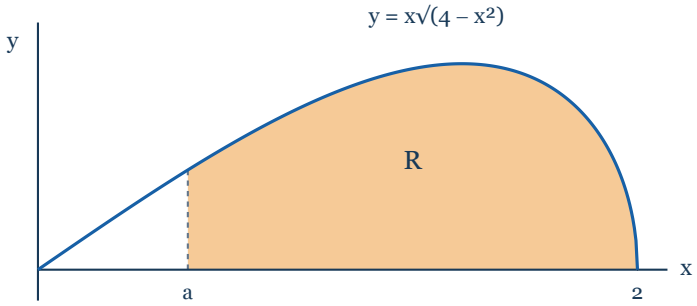
$$= \pi[4x^3/3 - x^5/5]_a^2$$

$$= \pi\big((32/3 - 32/5) - (4a^3/3 - a^5/5)\big)$$

$$= \frac{\pi}{15}(64 - 20a^3 + 3a^5).$$

$$V = 657\pi/160 \implies 3a^5 - 20a^3 + 77/32 = 0.$$

Try $a = 1$: $-467/32 \neq 0$. Try $a = 1/2$: the expression is zero. Thus $a - 1/2$ is a factor and $a = 1/2$. The volume strictly decreases on $0 < a < 2$, so this is the unique solution.



Exercise 5A — Core Pure Year 1 Pages 73–75 Solution 13

9. Rectangle R has length h and width r. Rotate it through $360°$ about the x-axis. Use integration to show that the cylinder has volume $V = \pi r^2 h$.

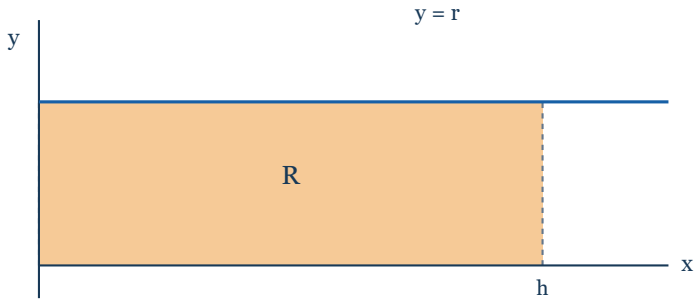
Working

The horizontal line has equation $y = r$.

$$V = \pi \int_0^h r^2 \, dx$$

$$= \pi [r^2 x]_0^h$$

$$= \pi r^2 h.$$



Prove the Volume of a Sphere Solution 14

Prove that the volume of a sphere is $V = \frac{4}{3}\pi r^3$.

Working

For the circle,

$$x^2 + y^2 = r^2, \quad y = \pm\sqrt{r^2 - x^2}.$$

Rotate the upper semicircle about the x-axis. First find one hemisphere:

$$V_{\text{half}} = \pi \int_0^r (\sqrt{r^2 - x^2})^2 \, dx$$

$$= \pi \int_0^r (r^2 - x^2) \, dx$$

$$= \pi [r^2 x - x^3/3]_0^r$$

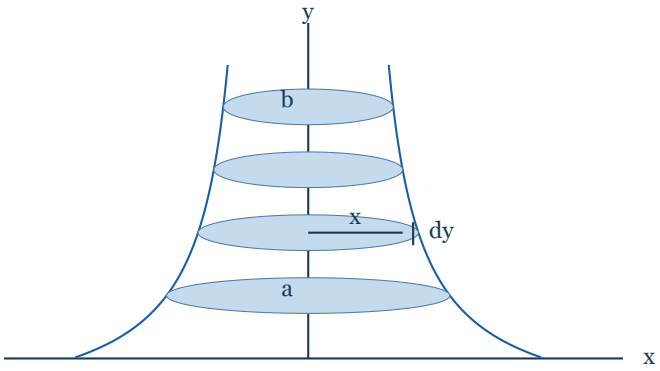
$$= \pi((r^3 - r^3/3) - 0) = \tfrac{2}{3}\pi r^3.$$

Double this:

$$V = 2\left(\tfrac{2}{3}\pi r^3\right) = \tfrac{4}{3}\pi r^3.$$

Revolving around the y-axis

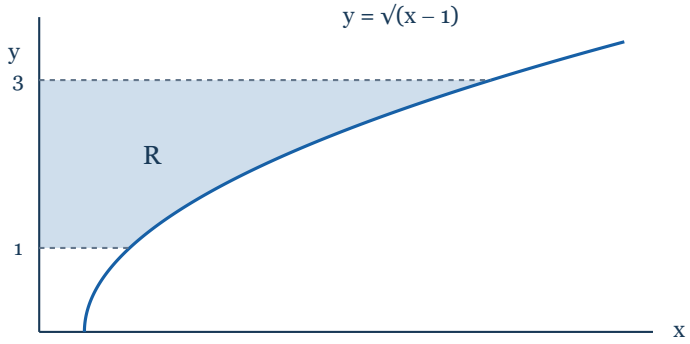
Solution 15



To revolve around the y-axis, swap the roles of the coordinates:

$$V = \pi \int_a^b x^2 \, dy.$$

[Textbook] The diagram shows the curve with equation $y = \sqrt{x-1}$. The region R is bounded by the curve, the y-axis and the lines $y = 1$ and $y = 3$. The region is rotated through 360° about the y-axis. Find the volume of the solid generated.



Working

$x = y^2 + 1.$

$$V = \pi \int_1^3 (y^2+1)^2 \, dy$$

$$= \pi \int_1^3 (y^4 + 2y^2 + 1) \, dy$$

$$= \pi [y^5/5 + 2y^3/3 + y]_1^3$$

$$= 1016\pi/15.$$

Test Your Understanding

Solution 16

[Textbook] A curve has equation $y = \sqrt[3]{2x+1}$. The region R is bounded by the curve, the y-axis and the lines $y = 2$ and $y = 4$. The region is rotated through 360° about the y-axis. Find the volume of the solid generated.

Working

$y^3 = 2x + 1 \implies x = \frac{1}{2}y^3 - \frac{1}{2}.$

$$V = \pi \int_2^4 (\tfrac{1}{2}y^3 - \tfrac{1}{2})^2 \, dy$$

$$= \frac{\pi}{4} \int_2^4 (y^6 - 2y^3 + 1) \, dy$$

$$= \frac{\pi}{4} [y^7/7 - y^4/2 + y]_2^4$$

$$= 7715\pi/14.$$

Exercise 5B — Core Pure Year 1 Pages 77–78 Solution 17

1. Find the exact volume when each curve is rotated through $360°$ about the y-axis between the given limits.

(a) $x = \frac{1}{2}y + 1,\ 2 \le y \le 5.$

a

$$V = \pi \int_2^5 (\tfrac{1}{2}y + 1)^2\,dy$$
$$= \pi \int_2^5 (y^2/4 + y + 1)\,dy$$
$$= \pi[y^3/12 + y^2/2 + y]_2^5$$
$$= \pi((125/12 + 25/2 + 5)$$
$$-(8/12 + 4/2 + 2))$$
$$= \pi(335/12 - 14/3) = 93\pi/4.$$

(b) $y = 2\sqrt{x},\ 0 \le y \le 1.$

b

$$y = 2\sqrt{x} \implies x = y^2/4.$$
$$V = \pi \int_0^1 (y^2/4)^2\,dy$$
$$= \pi \int_0^1 y^4/16\,dy$$
$$= \pi[y^5/80]_0^1 = \pi/80.$$

(c) $y = 1/x,\ 1 \le y \le 3.$

c

$$y = 1/x \implies x = 1/y.$$
$$V = \pi \int_1^3 (1/y)^2\,dy$$
$$= \pi \int_1^3 y^{-2}\,dy$$
$$= \pi[-1/y]_1^3$$
$$= \pi(-1/3 + 1) = 2\pi/3.$$

(d) $y = 2x^2 - 4,\ 5 \le x \le 11.$

d

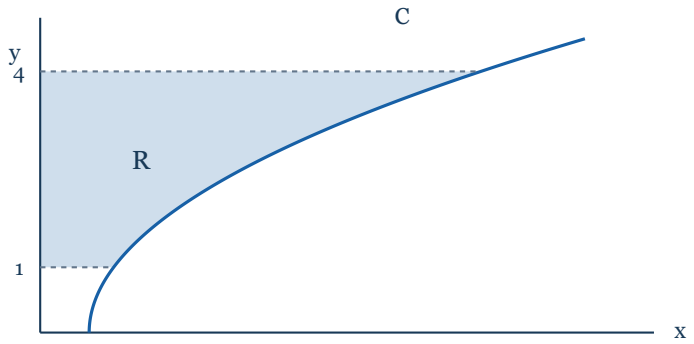
$$x^2 = \tfrac{1}{2}(y + 4).$$
$$x = 11 \implies y = 238; \quad x = 5 \implies y = 46.$$
$$V = \pi \int_{46}^{238} \tfrac{1}{2}(y + 4)\,dy$$
$$= \frac{\pi}{2}[y^2/2 + 4y]_{46}^{238}$$
$$= \frac{\pi}{2}((28322 + 952) - (1058 + 184))$$
$$= \frac{\pi}{2}(29274 - 1242) = 14016\pi.$$

Exercise 5B — Core Pure Year 1 Pages 77–78 Solution 18

2. $C:\ x = \frac{1}{2}y^2 + 1$. Region R is bounded by $y = 1$, $y = 4$, the y-axis and C. Rotate R through 2π radians about the y-axis. Find the volume.

Working

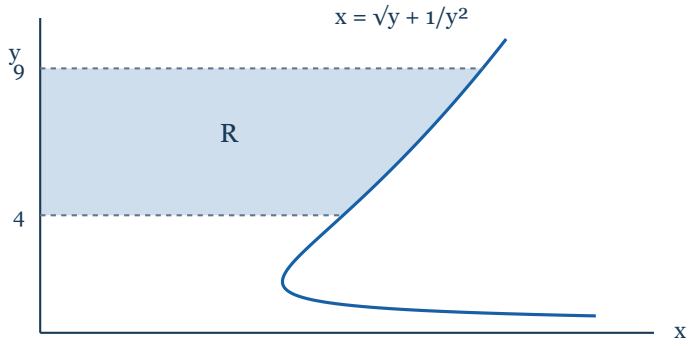
$$V = \pi \int_1^4 (\tfrac{1}{2}y^2 + 1)^2\,dy$$
$$= \pi \int_1^4 (y^4/4 + y^2 + 1)\,dy$$
$$= \pi[y^5/20 + y^3/3 + y]_1^4$$
$$= \pi((1024/20 + 64/3 + 4) - (1/20 + 1/3 + 1))$$
$$= \pi\left(\frac{3072 + 1280 + 240 - 3 - 20 - 60}{60}\right)$$
$$= 1503\pi/20.$$



Exercise 5B — Core Pure Year 1 Pages 77–78

Solution 19

3. Region R is bounded by $x = \sqrt{y} + y^{-2}$, $y = 4$, $y = 9$ and the y-axis. (a) Find its exact area. (b) Rotate R through 2π radians about the y-axis. Use integration to find the volume to 2 decimal places.



a

$$A = \int_4^9 (\sqrt{y} + y^{-2})\,dy$$
$$= \left[\tfrac{2}{3}y^{3/2} - 1/y\right]_4^9$$
$$= (18 - 1/9) - (16/3 - 1/4)$$
$$= \frac{648 - 4 - 192 + 9}{36} = \frac{461}{36}.$$

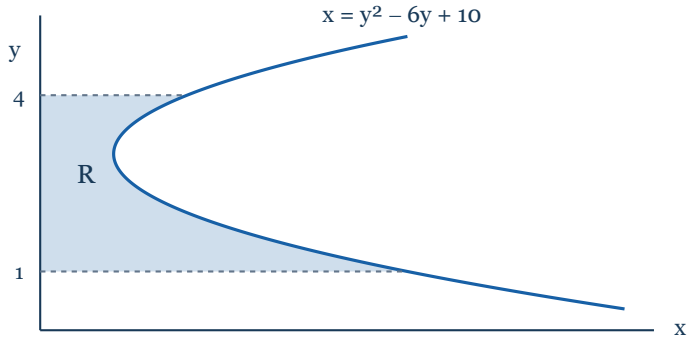
b

$$V = \pi \int_4^9 (\sqrt{y} + y^{-2})^2\,dy$$
$$= \pi \int_4^9 (y + 2y^{-3/2} + y^{-4})\,dy$$
$$= \pi[y^2/2 - 4y^{-1/2} - y^{-3}/3]_4^9$$
$$\approx \pi(39.1662 - 5.99479)$$
$$= 104.21 \quad (2\text{ d.p.}).$$

Exercise 5B — Core Pure Year 1 Pages 77–78

Solution 20

4. Region R is bounded by $x = y^2 - 6y + 10$, $y = 1$, $y = 4$ and the y-axis. (a) Find its area. (b) Rotate R through 360° about the y-axis; use integration to find the exact volume.



a

$$A = \int_1^4 (y^2 - 6y + 10)\,dy$$
$$= [y^3/3 - 3y^2 + 10y]_1^4$$
$$= (64/3 - 48 + 40) - (1/3 - 3 + 10) = 6.$$

b

$$V = \pi \int_1^4 (y^2 - 6y + 10)^2\,dy$$
$$= \pi \int_1^4 (y^4 - 12y^3 + 56y^2 - 120y + 100)\,dy$$
$$= \pi[y^5/5 - 3y^4 + 56y^3/3 - 60y^2 + 100y]_1^4$$
$$= \pi((1024/5 + 3584/3 - 1328)$$
$$- (1/5 + 56/3 + 37))$$
$$= 78\pi/5.$$

Exercise 5B — Core Pure Year 1 Pages 77–78 Solution 21

5. $C:\ y = 2x^2 + 5$. The region bounded by the y-axis, C and $y = 10$ is rotated through $360°$ about the y-axis. Find the exact volume.

Working

When $x = 0$, $y = 5$, so C cuts the axis at $(0, 5)$.

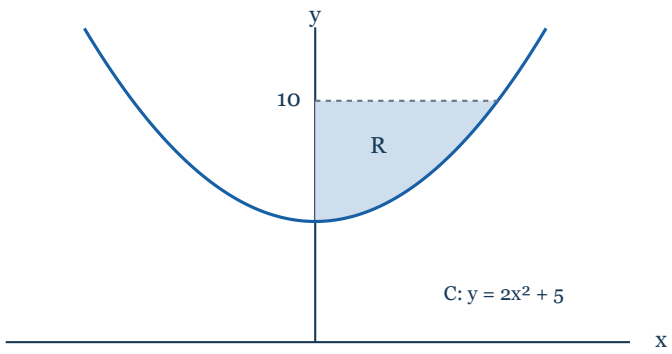
$x^2 = \frac{1}{2}(y - 5)$.

$$V = \pi \int_5^{10} \tfrac{1}{2}(y-5)\,dy$$

$$= \frac{\pi}{2}[y^2/2 - 5y]_5^{10}$$

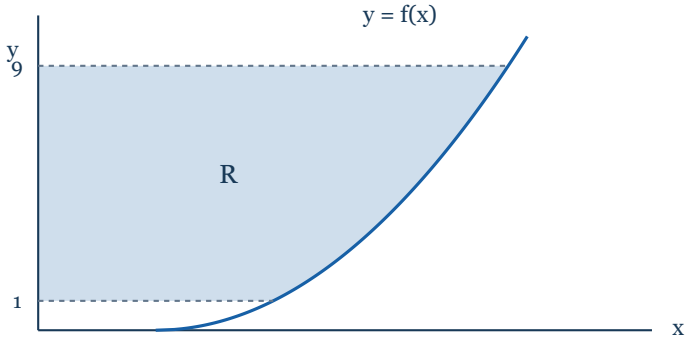
$$= \frac{\pi}{2}((50 - 50) - (25/2 - 25))$$

$$= 25\pi/4.$$



Exercise 5B — Core Pure Year 1 Pages 77–78 Solution 22

6. $f(x) = x^2 - 2x + 1,\ x \geq 1$. Region R is bounded by $y = f(x)$, the y-axis, $y = 1$ and $y = 9$. (a) Show that $x^2 = y + 2\sqrt{y} + 1$. (b) Rotate R through 2π radians about the y-axis; find the exact volume.



a

$y = x^2 - 2x + 1 = (x - 1)^2$.

Since $x \geq 1$,

$\sqrt{y} = x - 1 \implies x = \sqrt{y} + 1$.

$x^2 = (\sqrt{y} + 1)^2 = y + 2\sqrt{y} + 1$.

b

$$V = \pi \int_1^9 (y + 2\sqrt{y} + 1)\,dy$$

$$= \pi[y^2/2 + \tfrac{4}{3}y^{3/2} + y]_1^9$$

$$= \pi((81/2 + 36 + 9) - (1/2 + 4/3 + 1))$$

$$= 248\pi/3.$$

Exercise 5B — Core Pure Year 1 Pages 77–78 Solution 23

7. Region R is bounded by $y^3 + x^2 - 2y = 4$ and the x-axis. Rotate it about the y-axis. Find the exact volume.

Working

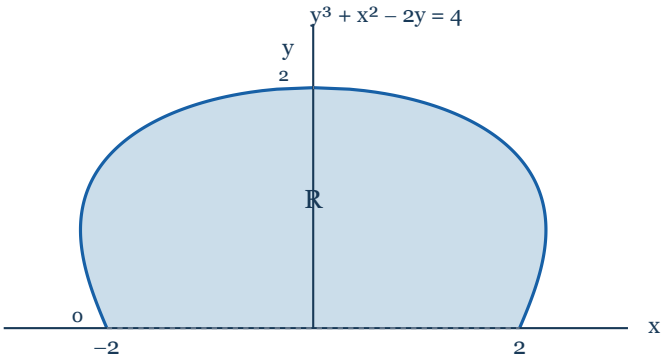
$y^3 + x^2 - 2y = 4 \implies x^2 = 4 + 2y - y^3.$

The upper intercept is $y = 2$.

$$V = \pi \int_0^2 (4 + 2y - y^3)\, dy$$

$$= \pi[4y + y^2 - y^4/4]_0^2$$

$$= \pi((8 + 4 - 4) - 0) = 8\pi.$$



Exercise 5B — Core Pure Year 1 Pages 77–78 Solution 24

8. $C:\ y^2 = 1/(2x + 1)$. Region R is bounded by C, the y-axis and $y = 4$. Rotate it through 2π radians about the y-axis. Find the volume.

Working

$2x + 1 = y^{-2} \implies x = \frac{1}{2}(y^{-2} - 1).$

$x^2 = \frac{1}{4}(y^{-4} - 2y^{-2} + 1).$

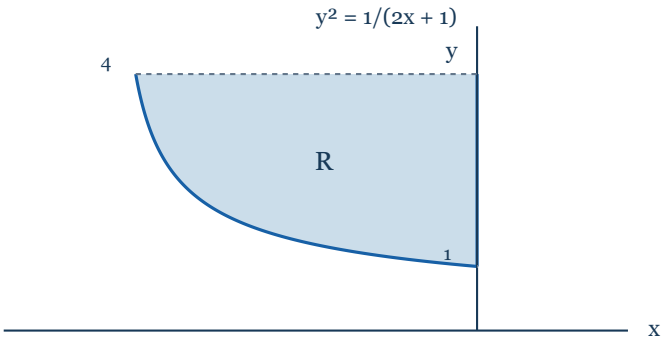
$$V = \frac{\pi}{4} \int_1^4 (y^{-4} - 2y^{-2} + 1)\, dy$$

$$= \frac{\pi}{4}[-1/(3y^3) + 2/y + y]_1^4$$

$$= \frac{\pi}{4}((-1/192 + 1/2 + 4) - (-1/3 + 2 + 1))$$

$$= \frac{\pi}{4}\left(\frac{-1 + 96 + 768 + 64 - 384 - 192}{192}\right)$$

$$= 117\pi/256 \approx 1.44 \quad (2\text{ d.p.}).$$



9. Right-angled triangle R has width r and height h. Rotate it through 2π radians about the y-axis. Use integration to show the cone has volume $V = \frac{1}{3}\pi r^2 h$.

Working

The hypotenuse has gradient h/r, so

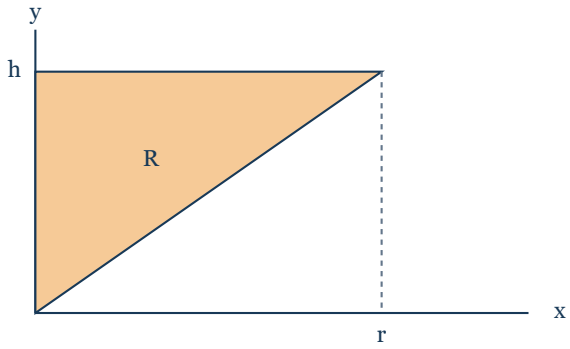
$$y = \frac{h}{r}x \implies x^2 = \frac{r^2}{h^2}y^2.$$

$$V = \pi \int_0^h \frac{r^2}{h^2} y^2 \, dy$$

$$= \frac{\pi r^2}{h^2}[y^3/3]_0^h$$

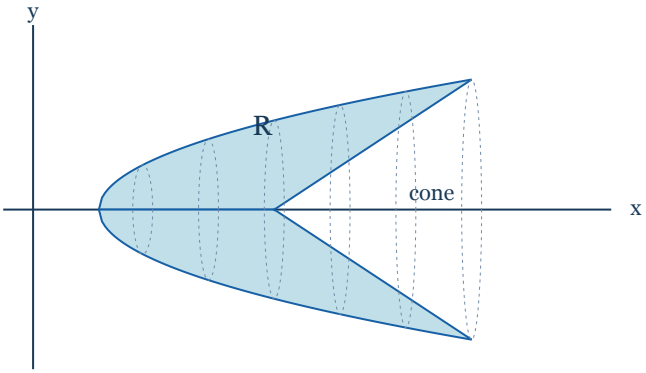
$$= \frac{\pi r^2}{h^2}(h^3/3 - 0)$$

$$= \tfrac{1}{3}\pi r^2 h.$$



Adding and Subtracting Volumes

Solution 26

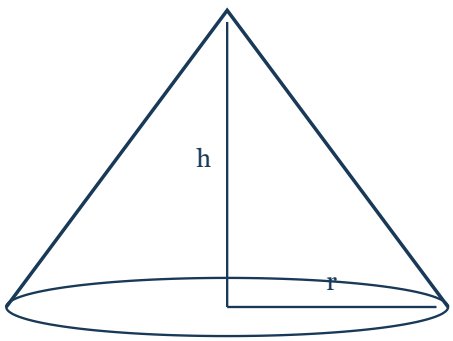


Suppose we revolve the shaded area about the x-axis. What strategy gives the volume?

Find the volume for the top curve. Then cut out (subtract) the cone.

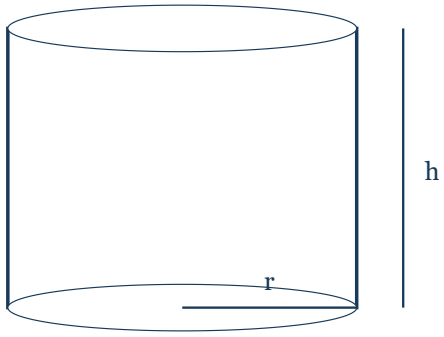
GCSE reminder: cone

$V = \frac{1}{3}\pi r^2 h.$



GCSE reminder: cylinder

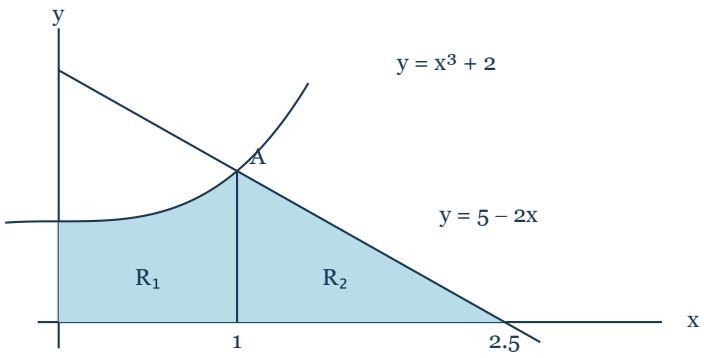
$V = \pi r^2 h.$



Example

Solution 27

The region R is bounded by $y = x^3 + 2$, $y = 5 - 2x$ and the coordinate axes. (a) Verify that the intersection A is $(1, 3)$. (b) Rotate R through 360° about the x-axis; find the volume.



a

$1^3 + 2 = 3, \qquad 5 - 2(1) = 3.$

Both equations pass through $(1, 3)$.

b

Find the two volumes separately.

$$V_1 = \pi \int_0^1 (x^3 + 2)^2 \, dx = 36\pi/7.$$

$5 - 2x$ meets the axis at $x = 2.5$. The second solid is a cone of radius 3 and height 1.5:

$V_2 = \frac{1}{3}\pi(3^2)(1.5) = 9\pi/2.$

$V = 36\pi/7 + 9\pi/2 = 135\pi/14.$

Volumes by Subtraction

Solution 28

Region R is bounded by $y = \sqrt{x}$, $y = 1/(8x)$ and $x = 1$. Rotate R through 360° about the x-axis. Find the exact volume.

Working

Find the volume under the top curve and subtract the volume under the bottom curve.

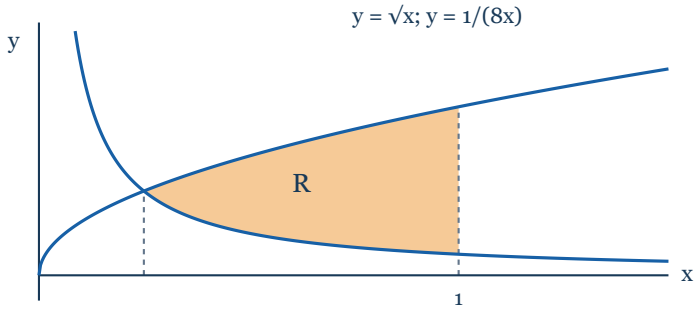
Intersection:

$$\frac{1}{8x} = \sqrt{x} \implies 8x^{3/2} = 1 \implies x = \tfrac{1}{4}.$$

$$V_1 = \pi \int_{1/4}^1 (\sqrt{x})^2 \, dx = 15\pi/32.$$

$$V_2 = \pi \int_{1/4}^1 (1/(8x))^2 \, dx = 3\pi/64.$$

$V_R = 15\pi/32 - 3\pi/64 = 27\pi/64.$



Test Your Understanding — Solution 29

The area between $y = x$ and $y = \sqrt[3]{x}$, where $x \geq 0$, is rotated through 360° about the x-axis. Find the volume.

Working

$x = x^{1/3} \implies x^3 = x \implies x^3 - x = 0.$

$x(x-1)(x+1) = 0 \implies x = -1, 0, 1.$

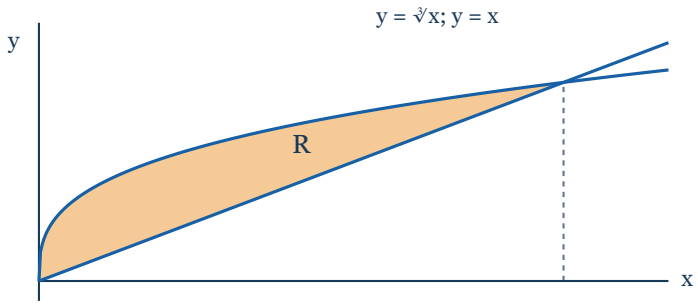
Use $0 \leq x \leq 1$.

$V_1 = \pi \int_0^1 (x^{1/3})^2 \, dx = 3\pi/5.$

The inner solid is a cone:

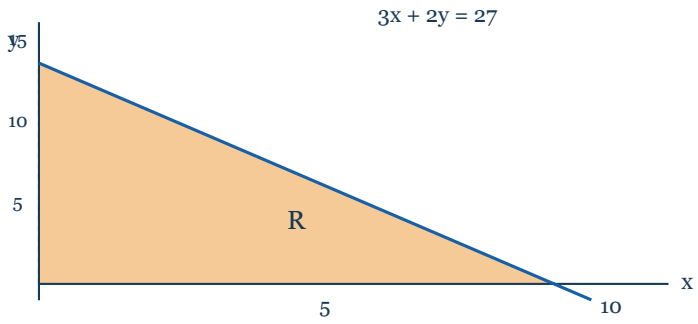
$V_2 = \frac{1}{3}\pi(1^2)(1) = \pi/3.$

$V_R = 3\pi/5 - \pi/3 = 4\pi/15.$



Exercise 5C — Core Pure Year 1 Pages 81–83 — Solution 30

1. The line is $3x + 2y = 27$. Use integration to find the volume of the region bounded by the line and the coordinate axes when rotated through 360°: (a) about the x-axis; (b) about the y-axis. (c) Check using the cone formula, stating each radius and height.



a

When $y = 0$, $x = 9$.

$y = \frac{1}{2}(27 - 3x).$

$y^2 = \frac{1}{4}(729 - 162x + 9x^2).$

$V = \frac{\pi}{4} \int_0^9 (729 - 162x + 9x^2) \, dx$

$= \frac{\pi}{4}[729x - 81x^2 + 3x^3]_0^9$

$= 2187\pi/4.$

b

When $x = 0$, $y = 13.5$.

$x = \frac{1}{3}(27 - 2y).$

$x^2 = \frac{1}{9}(729 - 108y + 4y^2).$

$V = \frac{\pi}{9} \int_0^{13.5} (729 - 108y + 4y^2) \, dy$

$= \frac{\pi}{9}[729y - 54y^2 + 4y^3/3]_0^{13.5}$

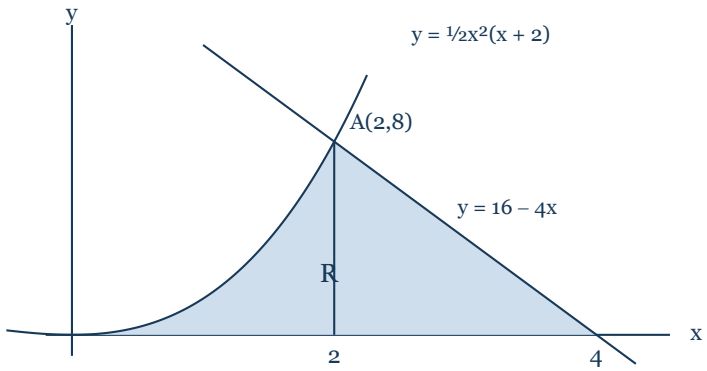
$= 729\pi/2.$

c

About the x-axis: $r = 13.5, h = 9$, so $V = \frac{1}{3}\pi(13.5)^2(9) = 2187\pi/4.$

About the y-axis: $r = 9, h = 13.5$, so $V = \frac{1}{3}\pi(9)^2(13.5) = 729\pi/2.$

2. Region R is bounded by $y = \frac{1}{2}x^2(x+2)$, $y = 16 - 4x$ and the x-axis. (a) Show that the intersection $A = (2, 8)$. (b) Rotate R through $360°$ about the x-axis; find the volume.



a

In the line, $x = 2$ gives $y = 16 - 8 = 8$. In the curve, $y = \frac{1}{2}(4)(2+2) = 8$. Thus $A = (2, 8)$.

b: curve

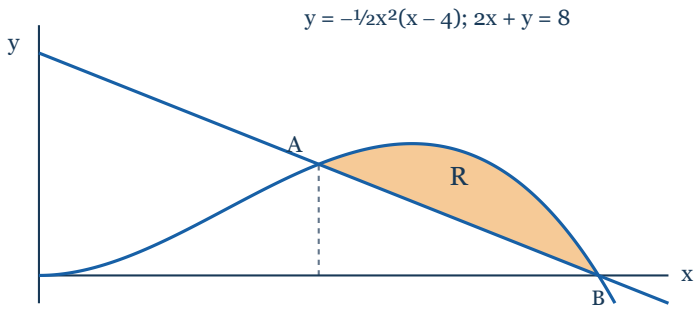
$$V_1 = \pi \int_0^2 (\tfrac{1}{2}x^2(x+2))^2\,dx$$
$$= \frac{\pi}{4}\int_0^2 x^4(x^2+4x+4)\,dx$$
$$= \frac{\pi}{4}\int_0^2 (x^6+4x^5+4x^4)\,dx$$
$$= \frac{\pi}{4}[x^7/7 + 2x^6/3 + 4x^5/5]_0^2$$
$$= 2272\pi/105.$$

b: total

The line meets $y = 0$ at $x = 4$. Its solid is a cone with $r = 8, h = 2$.

$$V_2 = \tfrac{1}{3}\pi(64)(2) = 128\pi/3.$$
$$V = 2272\pi/105 + 128\pi/3$$
$$= \frac{2272 + 4480}{105}\pi = 6752\pi/105.$$

3. Region R is bounded by $y = -\frac{1}{2}x^2(x-4)$ and $2x + y = 8$, as shown. (a) Show that $A = (2, 4)$, and give B. (b) Rotate R through $360°$ about the x-axis; find the volume.



a

At $x = 2$, the line gives $y = 8 - 4 = 4$; the curve gives $y = -\frac{1}{2}(4)(2-4) = 4$. Thus $A = (2, 4)$.

For $y = 0$, the line gives $x = 4$, so $B = (4, 0)$.

b: curve

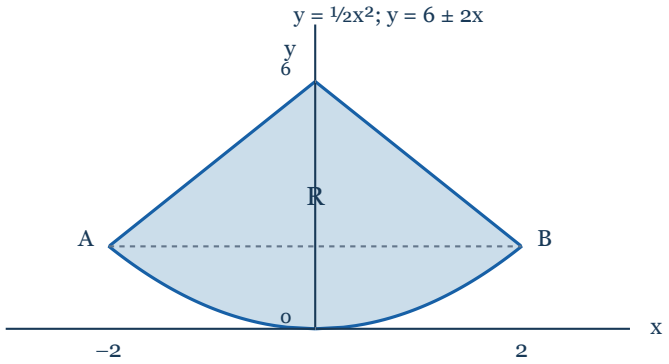
$$V_1 = \pi \int_2^4 (-\tfrac{1}{2}x^2(x-4))^2\,dx$$
$$= \frac{\pi}{4}\int_2^4 x^4(x^2-8x+16)\,dx$$
$$= \frac{\pi}{4}\int_2^4 (x^6-8x^5+16x^4)\,dx$$
$$= \frac{\pi}{4}[x^7/7 - 4x^6/3 + 16x^5/5]_2^4$$
$$= 3168\pi/105.$$

b: subtraction

The line generates a cone with $r = 4, h = 2$.

$$V_2 = \tfrac{1}{3}\pi(16)(2) = 32\pi/3.$$
$$V_R = 3168\pi/105 - 32\pi/3$$
$$= 2048\pi/105.$$

4. The shape is bounded by $y = \frac{1}{2}x^2$, $2x + y = 6$ and $2x - y = -6$. (a) Find A and B. (b) Rotate the shape about the y-axis; find the volume.



a

For the left line, $x = -2$ gives $y = 2$; the curve gives $y = \frac{1}{2}(-2)^2 = 2$. Thus $A = (-2, 2)$.

For the right line, $x = 2$ gives $y = 2$; the curve gives $y = \frac{1}{2}(2)^2 = 2$. Thus $B = (2, 2)$.

b

The curved lower part generates

$$V_1 = \pi \int_0^2 x^2\,dy = \pi \int_0^2 2y\,dy$$

$$= \pi[y^2]_0^2 = 4\pi.$$

The upper part is a cone with $r = 2, h = 4$:

$$V_2 = \tfrac{1}{3}\pi(4)(4) = 16\pi/3.$$

$$V = 4\pi + 16\pi/3 = 28\pi/3.$$

5. Region R is bounded by $x = -3$, $x = 3$, $y = 3$, the curve $x^2 + y^2 = 4$ and the x-axis, as shown. Rotate it about the y-axis. Find the volume.

Working

The curved cut-out generates

$$V_1 = \pi \int_0^2 (4 - y^2)\,dy$$

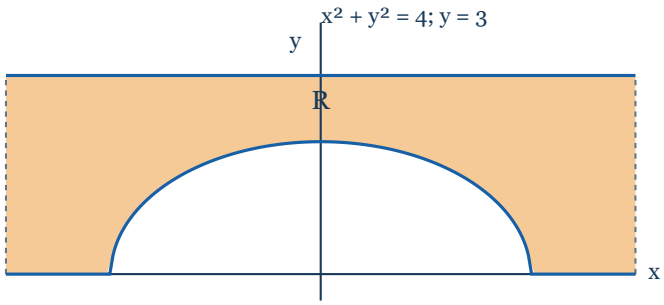
$$= \pi[4y - y^3/3]_0^2$$

$$= \pi(8 - 8/3) = 16\pi/3.$$

The outer cylinder has $r = 3, h = 3$:

$$V_2 = \pi(9)(3) = 27\pi.$$

$$V_R = 27\pi - 16\pi/3 = 65\pi/3.$$



6. Region R lies between $y = -\frac{1}{5}x^2 + 5$, the x-axis and the lines $y = 4 - x$, $y = 4 + x$, as shown. Rotate it about the y-axis. Find the volume.

Working

The curve meets the y-axis at $(0, 5)$, and $x^2 = 25 - 5y$.

$$V_1 = \pi \int_0^5 x^2\,dy = \pi \int_0^5 (25 - 5y)\,dy$$

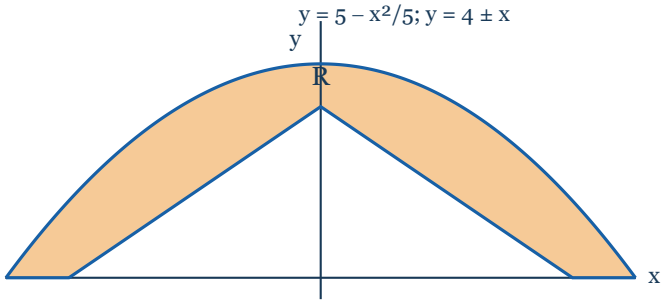
$$= \pi[25y - 5y^2/2]_0^5$$

$$= \pi(125 - 125/2) = 125\pi/2.$$

Subtract the cone, with $r = 4, h = 4$:

$$V_2 = \tfrac{1}{3}\pi(16)(4) = 64\pi/3.$$

$$V_R = 125\pi/2 - 64\pi/3 = 247\pi/6.$$



7. Region R is bounded by $6y^2 - x^3 + 4x = 0$, $x > 0$, the lines $4x - 3y = 4$, $4x + 3y = 4$, and $x = 4$, as shown. Rotate it about the x-axis. Find the exact volume.

Working

At $y = 0$:

$$4x - x^3 = x(4 - x^2) = 0.$$

Since $x > 0$, the curve meets the axis at $(2, 0)$.

$$V_1 = \pi \int_2^4 y^2\,dx = \frac{\pi}{6} \int_2^4 (x^3 - 4x)\,dx$$

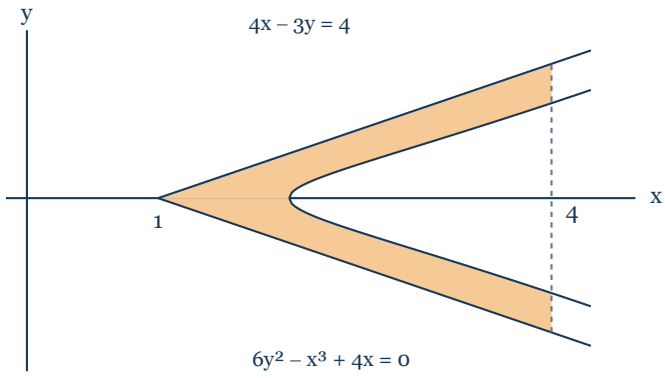
$$= \frac{\pi}{6}[x^4/4 - 2x^2]_2^4$$

$$= \frac{\pi}{6}((64 - 32) - (4 - 8)) = 6\pi.$$

The outer cone has $r = 4, h = 3$:

$$V_2 = \tfrac{1}{3}\pi(16)(3) = 16\pi.$$

$$V_R = 16\pi - 6\pi = 10\pi.$$



Exercise 5C — Core Pure Year 1 Pages 81–83 Solution 37

8. Region R is bounded by $y = 4 - x^2$, $y = \sqrt[3]{x}$, the y-axis and $x = 1$. Rotate it through 360° about the x-axis. Find the exact volume.

Working

The upper curve generates

$$V_1 = \pi \int_0^1 (4 - x^2)^2 \, dx$$

$$= \pi \int_0^1 (16 - 8x^2 + x^4) \, dx$$

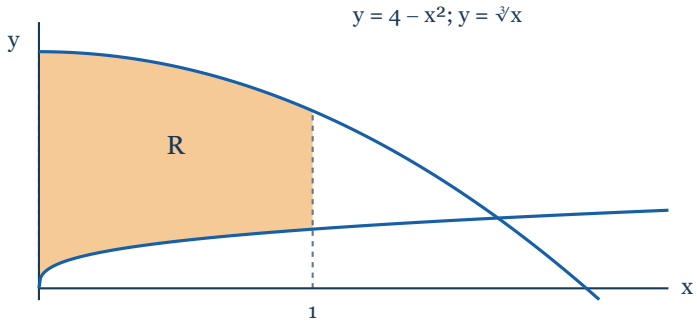
$$= \pi[16x - 8x^3/3 + x^5/5]_0^1$$

$$= \pi(16 - 8/3 + 1/5) = 203\pi/15.$$

The lower curve generates

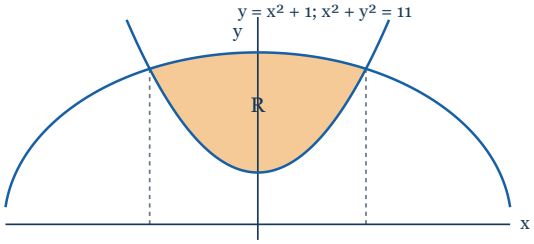
$$V_2 = \pi \int_0^1 x^{2/3} \, dx = \pi[3x^{5/3}/5]_0^1 = 3\pi/5.$$

$$V_R = 203\pi/15 - 3\pi/5 = 194\pi/15.$$



Exercise 5C — Core Pure Year 1 Pages 81–83 Solution 38

9. Region R is bounded by $y = x^2 + 1$ and $x^2 + y^2 = 11$, as shown. (a) Find the x-coordinates of their intersections. (b) Rotate R through 360° about the x-axis. Find the volume to 2 decimal places.



a

$$x^2 + (x^2 + 1)^2 = 11$$

$$x^4 + 3x^2 - 10 = 0$$

$$(x^2 - 2)(x^2 + 5) = 0.$$

$$x = \pm\sqrt{2}.$$

b: circle

The circle generates

$$V_1 = \pi \int_{-\sqrt{2}}^{\sqrt{2}} (11 - x^2) \, dx$$

$$= \pi[11x - x^3/3]_{-\sqrt{2}}^{\sqrt{2}}$$

$$= \frac{\pi}{3}((33\sqrt{2} - 2\sqrt{2})$$

$$- (-33\sqrt{2} + 2\sqrt{2}))$$

$$= 62\sqrt{2}\pi/3.$$

b: result

The parabola generates

$$V_2 = \pi \int_{-\sqrt{2}}^{\sqrt{2}} (x^2 + 1)^2 \, dx$$

$$= \pi \int_{-\sqrt{2}}^{\sqrt{2}} (x^4 + 2x^2 + 1) \, dx$$

$$= \pi[x^5/5 + 2x^3/3 + x]_{-\sqrt{2}}^{\sqrt{2}}$$

$$= \pi((4\sqrt{2}/5 + 4\sqrt{2}/3 + \sqrt{2})$$

$$- (-4\sqrt{2}/5 - 4\sqrt{2}/3 - \sqrt{2}))$$

$$= \pi(8\sqrt{2}/5 + 8\sqrt{2}/3 + 2\sqrt{2})$$

$$= 94\sqrt{2}\pi/15.$$

$$V_R = 62\sqrt{2}\pi/3 - 94\sqrt{2}\pi/15$$

$$= 72\sqrt{2}\pi/5 = 63.98 \quad (2 \text{ d.p.}).$$

Modelling

Solution 39



In the 1990 film Ghost, Patrick Swayze's character returns as a ghost to resolve unfinished business. In a famous scene he makes pottery with his fiancée, played by Demi Moore.

The filmmakers want to know how much clay to buy. Model the outside curve by

$x = \frac{1}{100}(y-30)^2 + 10,$

with x, y in cm. The pottery spins about the y-axis and is 40 cm high. Find the volume to 3 significant figures, treating the model as solid.

Volume

$$V = \pi \int_0^{40} \left(\tfrac{1}{100}(y-30)^2 + 10\right)^2 \, dy$$

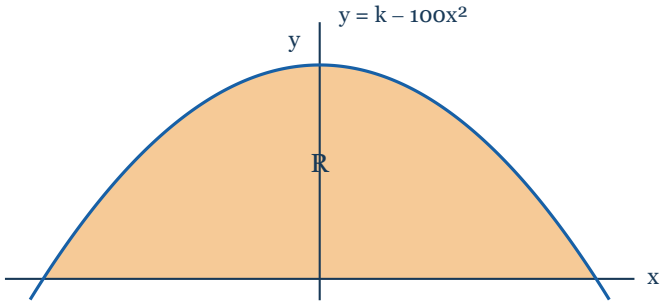
$$= \pi \left[\frac{(y-30)^5}{50000} + \frac{(y-30)^3}{15} + 100y \right]_0^{40}$$

$$= \frac{19064\pi}{3} \approx 20000 \text{ cm}^3 \quad (3 \text{ s.f.}).$$

Test Your Understanding

Solution 40

[Textbook] A manufacturer wants to cast a prototype for a new design for a pen barrel out of solid resin. The shaded region shown in the diagram is used as a model for the cross-section of the pen barrel. The region is bounded by the x-axis and the curve with equation $y = k - 100x^2$, and will be rotated around the y-axis. Each unit on the coordinate axes represents 1cm.



(a) Suggest a suitable value for k. (Let's say pens are 10cm long)

(b) Use your value of k to estimate the volume of resin needed to make the prototype.

(c) State one limitation of this model.

a Choose $k = 10$: pens are around 10–15 cm long.

c
The cross-section of the pen is unlikely to match the curve exactly.

b

$$y = 10 - 100x^2 \implies x = \sqrt{(10-y)/100}.$$

$$V = \pi \int_0^{10} \left(\sqrt{\frac{10-y}{100}} \right)^2 \, dy$$

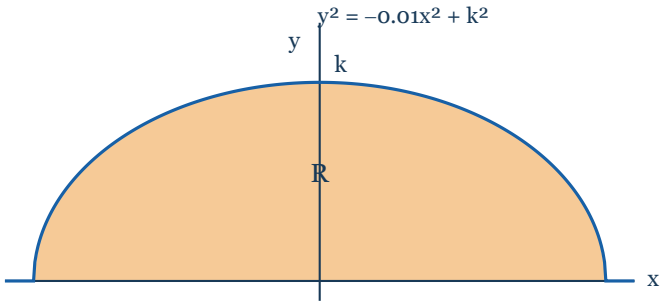
$$= \pi \int_0^{10} \frac{10-y}{100} \, dy$$

$$= \frac{\pi}{100} [10y - y^2/2]_0^{10}$$

$$= \pi/2 \approx 1.57 \text{ cm}^3.$$

Exercise 5D — Core Pure Year 1 Pages 84–86 Solution 41

1. The outside of a large tent at a fair is modelled by $y^2 = -0.01x^2 + k^2$, with units in metres. Rotate the upper outline about the y-axis. (a) Suggest a suitable k. (b) Use it to estimate the capacity. (c) State one limitation.



a

k is the central height in metres; for example, $5 \le k \le 15$. Choose $k = 10$.

b

$$V = \pi \int_0^k x^2\,dy = 100\pi \int_0^k (k^2 - y^2)\,dy$$

$$= 100\pi[k^2 y - y^3/3]_0^k$$

$$= 100\pi(2k^3/3) = 200k^3\pi/3 \text{ m}^3.$$

For $k = 10$, $V = 200000\pi/3 \approx 209440 \text{ m}^3$.

c

The tent may not follow this exact profile, and may not be exactly circular when viewed from above.

Exercise 5D — Core Pure Year 1 Pages 84–86 Solution 42

2. Half the outline of a rugby ball is modelled by $y^2 = 4(16 - x)$, with measurements in cm. Rotate the curve through 360° about the y-axis. Find the total volume.

Working

$$y^2 = 4(16 - x) \implies x = 16 - y^2/4.$$

$$V = \pi \int_{-8}^{8} (16 - y^2/4)^2\,dy$$

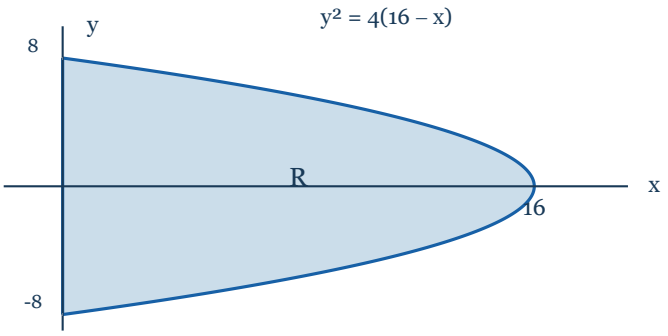
$$= \pi \int_{-8}^{8} (256 - 8y^2 + y^4/16)\,dy$$

$$= \pi[256y - 8y^3/3 + y^5/80]_{-8}^{8}$$

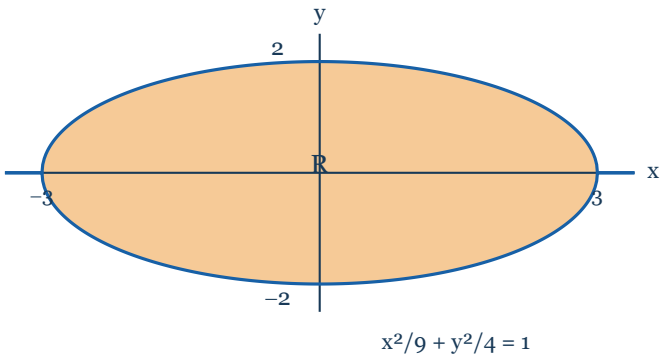
$$= 2\pi(2048 - 4096/3 + 32768/80)$$

$$= 4096\pi(1 - 2/3 + 1/5)$$

$$= 4096\pi(8/15) = 32768\pi/15 \text{ cm}^3.$$



3. Model an egg cross-section by $x^2/9 + y^2/4 = 1$, with dimensions in cm. (a) Calculate the volume formed by rotation through $360°$ about the x-axis. (b) Calculate the volume for rotation about the y-axis. (c) Which solid resembles an egg more closely?



a

$y^2 = 4(1 - x^2/9).$

$V = 4\pi \int_{-3}^{3} (1 - x^2/9)\,dx$

$= 4\pi[x - x^3/27]_{-3}^{3}$

$= 4\pi((3 - 1) - (-3 + 1)) = 16\pi \text{ cm}^3.$

b

$x^2 = 9(1 - y^2/4).$

$V = 9\pi \int_{-2}^{2} (1 - y^2/4)\,dy$

$= 9\pi[y - y^3/12]_{-2}^{2}$

$= 9\pi((2 - 2/3) - (-2 + 2/3)) = 24\pi \text{ cm}^3.$

c

Rotation about the x-axis gives the more egg-like shape. Rotation about the y-axis gives a flatter, disc-like shape.

4. An egg timer is 16 cm high and modelled as a solid of revolution about the y-axis. Its upper curve is $x = \sqrt[3]{y}$, with a matching lower half. Sand flows at 8 cm^3/min. The timer must empty in 5 minutes. Find, to 2 decimal places, the initial height of sand in the top half.

Working

The sand volume is $8 \times 5 = 40 \text{ cm}^3$. Let b be its height.

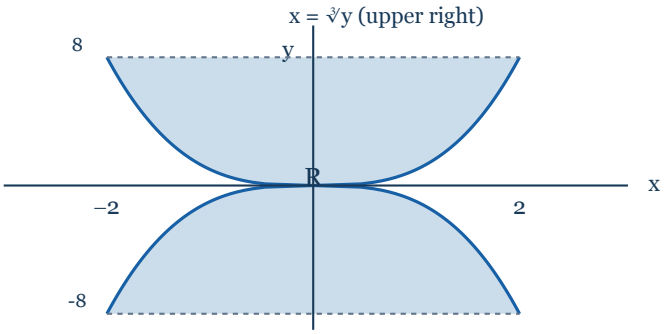
$40 = \pi \int_{0}^{b} (\sqrt[3]{y})^2\,dy$

$40 = \pi[3y^{5/3}/5]_{0}^{b}$

$40 = \frac{3}{5}\pi b^{5/3}$

$b^{5/3} = \dfrac{200}{3\pi}$

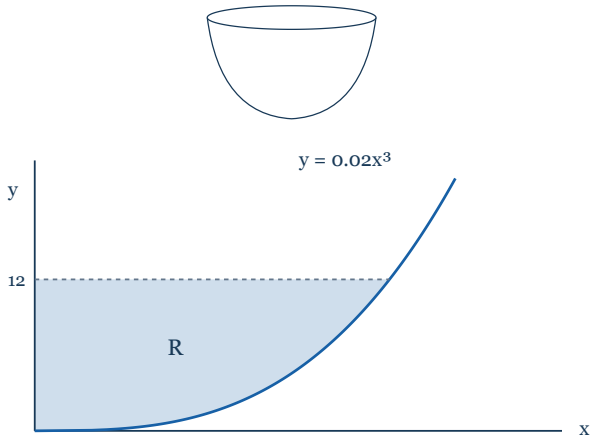
$b = \left(\dfrac{200}{3\pi}\right)^{3/5} = 6.25 \text{ cm}.$



Exercise 5D — Core Pure Year 1 Pages 84–86 — Solution 45

5. An electric stand mixer bowl is 18 cm high. Model it by rotating $y = 0.02x^3$ through 2π radians about the y-axis. (a) Find the diameter. (b) Find the maximum liquid volume.

A 12 cm-high paddle just touches the bowl and forms region R, as shown. (c) Find its area. (d) The paddle rotates about the y-axis: find the proportion of the bowl's volume it can mix.



a

$18 = 0.02x^3 \implies x^3 = 900.$

$d = 2\sqrt[3]{900} = 19.3$ cm.

b

$$V = \pi \int_0^{18} (\sqrt[3]{50y})^2 \, dy$$
$$= \sqrt[3]{2500}\pi[3y^{5/3}/5]_0^{18}$$
$$= \sqrt[3]{2500}\pi(3 \cdot 18^{5/3}/5)$$
$$= 3162.8 \text{ cm}^3.$$

c

$$A_R = \int_0^{12} \sqrt[3]{50y} \, dy$$
$$= \sqrt[3]{50}[3y^{4/3}/4]_0^{12}$$
$$= 75.9 \text{ cm}^2.$$

d

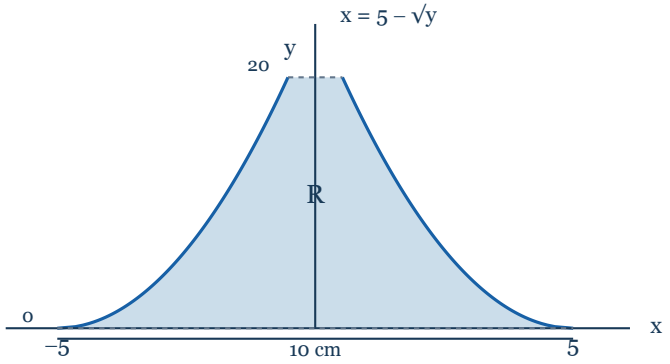
$$V_p = \pi \int_0^{12} (\sqrt[3]{50y})^2 \, dy$$
$$= \sqrt[3]{2500}\pi[3y^{5/3}/5]_0^{12}$$
$$= \sqrt[3]{2500}\pi(3 \cdot 12^{5/3}/5)$$
$$= 1609.1 \text{ cm}^3.$$

Proportion $= 1609.1/3162.8 = 0.509$.

Directly: $(12/18)^{5/3} = (2/3)^{5/3} = 0.509$.

Exercise 5D — Core Pure Year 1 Pages 84–86 — Solution 46

6. A vase has base width 10 cm and height 20 cm. Model its edge by $x = 5 - \sqrt{y}$, rotating about the y-axis through 360°. (a) Estimate its capacity. (b) It initially contains water to height 10 cm. Flowers displace 50 cm³ of water. Will it overflow?



a

$$V = \pi \int_0^{20} (5 - \sqrt{y})^2 \, dy$$
$$= \pi \int_0^{20} (25 - 10\sqrt{y} + y) \, dy$$
$$= \pi[25y - \tfrac{20}{3}y^{3/2} + y^2/2]_0^{20}$$
$$= \pi(500 - \tfrac{20}{3} \cdot 20^{3/2} + 200)$$
$$= 325.8 \text{ cm}^3.$$

b

The water initially occupies

$$V_{10} = \pi[25y - \tfrac{20}{3}y^{3/2} + y^2/2]_0^{10}$$
$$= \pi(250 - \tfrac{20}{3} \cdot 10^{3/2} + 50)$$
$$= 280.2 \text{ cm}^3.$$

Adding 50 cm³ gives 330.2 cm³, exceeding 325.8 cm³. It will overflow.

7. A solid wooden spinning top is 18 cm wide and 24 cm high. Its cross-section uses part of $y^2 = 4(x+9)$, the line $y = 2x + 18$, and symmetry about the y-axis. Rotate it about the y-axis. Find the total wood volume.

Working

The line meets the axes at $(-9, 0)$ and $(0, 18)$. The upper cone has $r = 9, h = 18$:

$V_1 = \frac{1}{3}\pi(81)(18) = 486\pi \text{ cm}^3$.

The curve meets the lower axis at $(0, -6)$:

$$V_2 = \pi \int_{-6}^{0} (y^2/4 - 9)^2 \, dy$$

$$= \pi \int_{-6}^{0} (y^4/16 - 9y^2/2 + 81) \, dy$$

$$= \pi[y^5/80 - 3y^3/2 + 81y]_{-6}^{0}$$

$$= \pi(0 - (-486/5 + 324 - 486))$$

$$= \pi(486 - 1620 + 2430)/5 = 1296\pi/5.$$

$V = 486\pi + 259.2\pi = 745.2\pi \text{ cm}^3$.

